CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

PROPOSAL PRESENTATION

Planner Copilot

A local AI assistant for production planning simulations

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Team 1 · Guided by Dr. Hamidreza Heidari · 20 October 2026

THE PROBLEM

Good production planning needs simulation, and experts

A busy week, missing workers or a broken machine: a simulation shows what will happen before it does.

Today mostly only a simulation expert can run it correctly, and small companies often have none.

An AI chatbot alone cannot plan production, and nobody can check its answer without an expert.

Key motivations

- Small companies get answers without a simulation expert
- Large companies save expert time
- Company data stays in-house
- Reliable AI help for planning is needed now

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PROJECT SCOPE

What we will and will not do

Aim

Test whether a local AI assistant with code checks can answer planning questions at the level of published expert simulations.

Includes

- About 4 published simulation cases
- Test questions with correct answers
- A local AI assistant with code checks
- Comparison of three setups

Out of scope (current phase)

- A company’s own production data and models
- Commercial simulation tools
- Use in a real factory

GOALS

Three measurable targets

≥ 90%

Close to the expert

For questions with an answer, the AI’s answer is close to the expert’s answer.

≤ 2%

Confidently wrong

The AI gives a wrong answer and sounds sure. This is the most dangerous error.

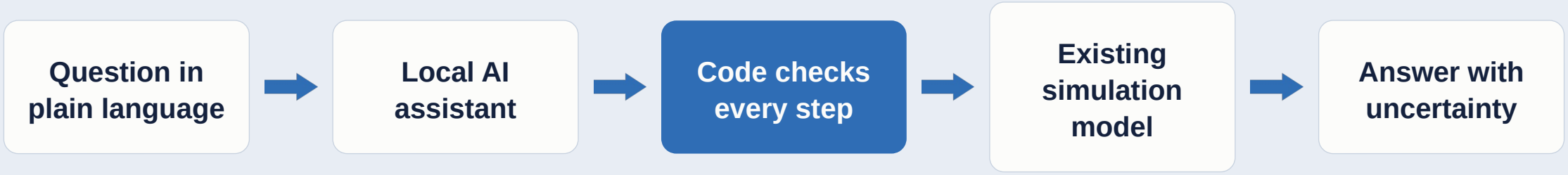
≥ 90%

Correct refusals

When a question cannot be answered, the AI says so instead of guessing.

OUR IDEA

An AI assistant that runs the simulation, and checks itself



Example: “Busy week, +20% orders. Do we need a fourth worker?”

TECHNOLOGY STACK

Process	Tool	Purpose
Simulation	JaamSim, Python	A computer model of a factory: shows what happens if orders, workers or machines change
AI model	Open-source LLM (Ollama)	The AI that understands plain-language questions; runs on our own computer
Code checks	Python, MCP server	Checks every AI step and refuses unclear questions
Interface	Gradio	A simple web page to ask questions and see charts
Compute	Google Colab GPU	A cloud computer with a graphics card to run the AI
Code sharing	GitHub	A shared place to store and track our code

Tools may change as the project progresses !

MVP · WHAT ALREADY WORKS

Three real answers from our first version

① Normal question → answer

“Normal week, current staffing: how many hours after the Friday shift until all orders are done?” → 3.37 h. Expert result: 3.31 h ✓

② Outside the validated range → warning

“+20 % orders and only 2 workers?” → about 38 h, with the warning: outside the validated range (3–4 workers), indicative only ✓

③ Outside the model → refusal

“What will the overtime cost in euros?” → Refused: the model has no cost data; it answers overtime and unfinished orders ✓

Behind every answer

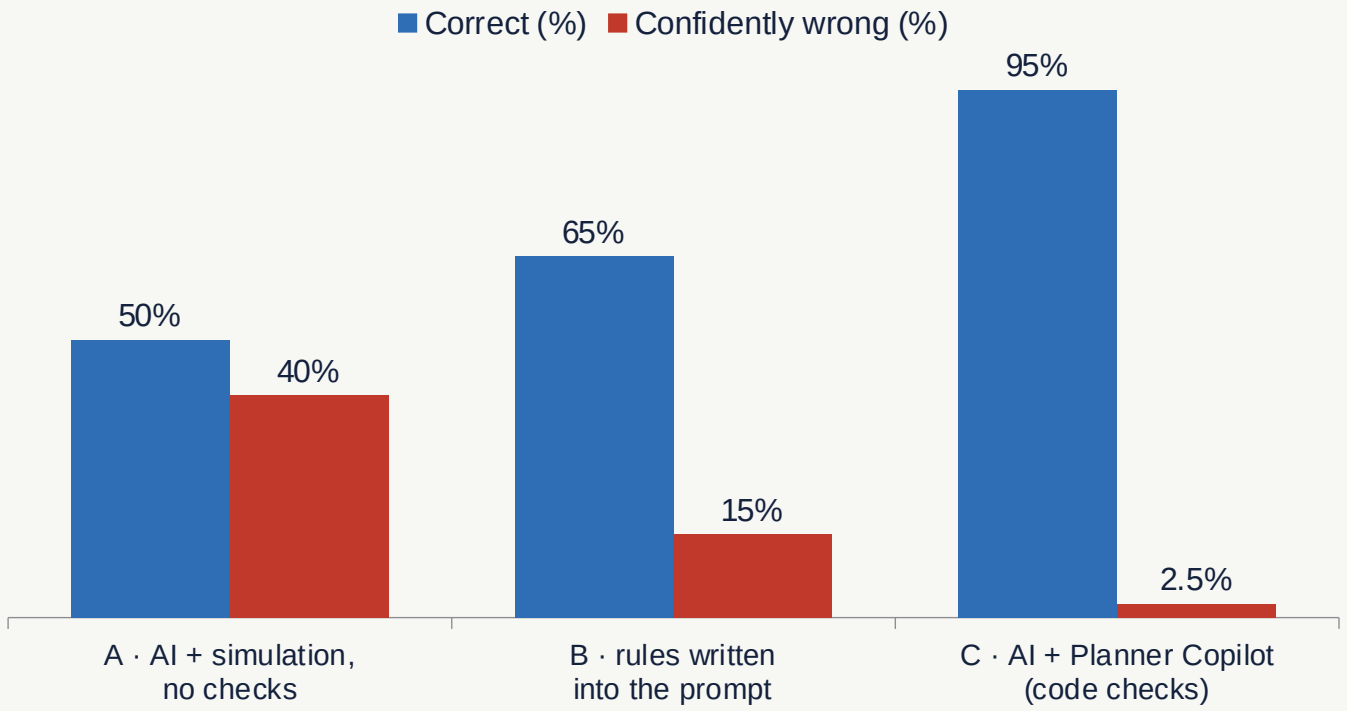
Case card → code checks (scope, units, validated range, replications) → published simulation model. Local open-source AI model (Qwen 2.5, 7B).

Case 2 (SME job shop) · development run, 17 Oct 2026 · every number comes from the simulation, not from the AI

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MVP · FIRST FEASIBILITY TEST

Code checks make the difference



Confidently wrong answers: 16 → 1 out of 40.

Rules written into the prompt help a little; checks in code help much more.

Small test: 40 questions, 2 cases, one local 7B model (16 Oct 2026).

Full experiments: about 150 questions, 4 cases, several models (P4, 8 Dec).

Same 40 questions for all three setups · each answer graded correct / cautious / confidently wrong · details: github.com/bodhi584/planner-copilot

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PLAN

Four steps to get there

1 · Find cases

About 4 published simulation cases from different industries, using public data only.

2 · Rebuild and ask

Rebuild each case and write test questions with the correct answers.

3 · Build the assistant

A local AI assistant with code checks, built with open-source tools.

4 · Compare

Three setups: expert, AI + simulation, and AI + Planner Copilot + simulation.

TEAM

Who does what

Yu Han

Team lead

Case 2
Code checks (guardrail engine)

Melisa Subasi

Questions & report

Case 1
Test questions
Final report editor

Meron Hagos

Interface & literature

Case 3
User interface
Literature review

Alejandro Rivas

Experiments

Case 4
Experiments
GitHub & demo

A case is one published factory simulation model with known expert results.

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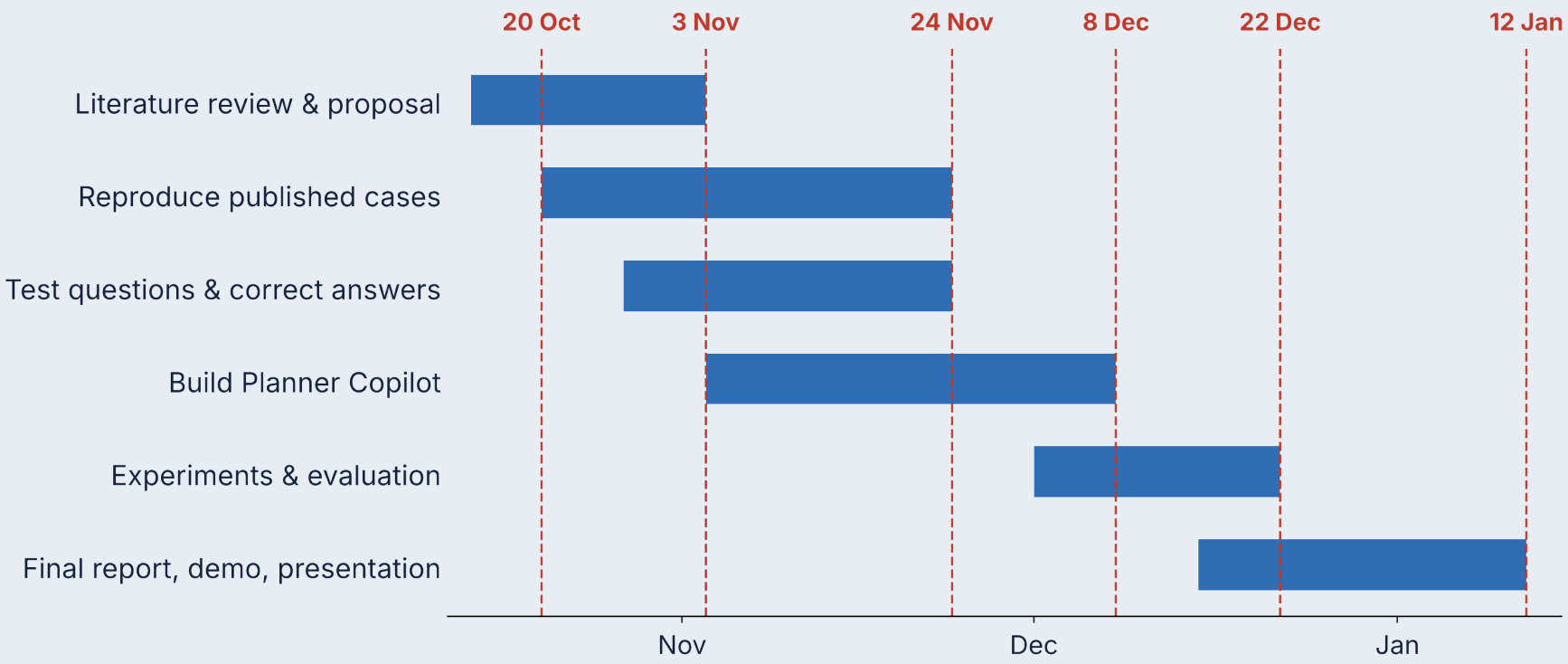
TIMELINE

From proposal to final hand-in

Date	Milestone
20 Oct	Proposal, team, GitHub
3 Nov	Literature report, method outline, baseline simulation
24 Nov	Cases rebuilt, first version of the assistant
8 Dec	First experiments
22 Dec	Demo, experiments and analysis, report draft
12 Jan	Final presentation, report, demo video

GANTT CHART

Project schedule



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EXPECTED OUTCOMES

What we will hand in

Test questions

Questions with correct answers for about 4 published cases.

Planner Copilot

A working open-source assistant that runs the simulations.

Comparison

Measured results for three setups against our targets.

Report and demo

Final report, demo video and a GitHub repository.

BILL OF MATERIALS

What the project needs

Component	Specification	Cost
GPU access (Google Colab)	Cloud GPU for the local AI	Free tier, to be confirmed
Simulation cases and data	Published, public	Free
Software	Open source	Free

RISKS, ETHICS

Keeping it safe and realistic

Ethics

Only public data, so no privacy issue. All sources are cited.

Main risk

Rebuilding the cases may take longer than planned. Fallback: fewer cases.

Thank You !

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github.com/bodhi584/planner-copilot